Final Project Report MATH 563

Alexandre St-Aubin
Student ID: 261115785

Jonathan Campana
Student ID: 260985613

Edmund Wu
Student ID: 261033501

Elliot Forcier-Poirier
Student ID:

Abstract

give some background

1 Introduction

1.1 Notation

Throughout this paper, we will denote the *proximal operator* of f with parameter λ by

$$\mathbf{prox}_{\lambda f}(v) := \arg\min_{x} \left\{ f(x) + \frac{1}{2\lambda} \|x - v\|_2^2 \right\} \tag{1}$$

The *indicator function* for a convex set C will be defined as

$$\delta_C(x) := \begin{cases} 0, & \text{if } x \in C \\ +\infty, & \text{if } x \notin C \end{cases} \tag{2}$$

We also denote the *Euclidean Projection* on a set C by

$$\mathbf{P}_C(v) := \arg\min_{x \in C} \|x - v\|_2 \tag{3}$$

1.2 Preleminaries

We first note an important relation between the proximal and subdifferential operators.

Proposition 1. Given a function $f \in \Gamma$, we have

$$\mathbf{prox}_{\lambda f} = (I + \lambda \partial f)^{-1}, \tag{4}$$

where $(I + \lambda \partial f)^{-1}$, for $\lambda > 0$, is called the **resolvent** of the operator ∂f. In particular, the proximal operator is the resolvent of the subdifferential operator [3].

Definition 1 (Maximal Monotone operator)**.** A monotone operator is **maximal** if its graph is not properly contained in the graph of any other monotone operator. If A is monotone (resp. maximal monotone) then so are A^{-1} and λA if $\lambda > 0$.

The next proposition, given in [2], will be useful in the construction of the optimizing algorithms.

Proposition 2. If F is maximal monotone, then $(I + \lambda F)^{-1}(\hat{x})$ exists and is unique for all $\hat{x} \in R^n$. The value $x = (I + \lambda F)^{-1}(\hat{x})$ of the resolvent is the unique solution of the monotone inclusion

$$\hat{x} \in x + \lambda F(x) \tag{5}$$

Preprint. Under review.

Proposition 3 (Proximal operator splitting)**.** If f is separable across two variables, so $f(x,y) = \varphi(x) + \psi(y)$, then

$$\mathbf{prox}_f(x,y) = (\mathbf{prox}_\varphi(x), \mathbf{prox}_\psi(y)) \tag{6}$$

Thus, evaluating the proximal operator of a separable function reduces to evaluating the proximal operators for each of the separable parts, which can be done independently [3].

Proposition 4. (Proximal operator of norms) If $f = \|\cdot\|$ is a general norm, then f^* is the indicator function on the unit ball for the dual norm $\|\cdot\|_*$, namely

$$\|\cdot\|^* = \delta_{\|\cdot\|_* \le 1}$$

where $\|\cdot\|_*$ is given by

$$\|v\|_* = \sup_x \{\langle v, x\rangle \mid \|x\| \le 1\}$$

In particular, the *Moreau Decomposition Theorem* gives that

$$\mathbf{prox}_{\|\cdot\|}(v) = \mathbf{P}_{\{x\,:\,\|x\|_* \le 1\}}(v) - v$$

1.3 Derivations of the proximal maps

We provide a quick proof of the proximal operators of the L1 norm, the L2 norm, the iso norm, and the box prox. Let $b \in \mathbb{R}^d$ be fixed.

Let $g : y \mapsto t\|y-b\|_1$. We have

$$\mathbf{prox}_g(y) := \arg\min_u \{\, t\|u-b\|_1 + \frac{1}{2}\|y-u\|_2^2 \,\} = \begin{cases} b & y-b \in [-t,t] \\ y - \mathrm{sign}(y-b)\cdot t & \text{otherwise} \end{cases}$$

We consider first-order optimality conditions, with the following cases:

- If $u-b > 0$, then
$$t - y + u = 0 \quad \Leftrightarrow \quad u = y - t$$
- If $u-b < 0$, then
$$-t - y + u = 0 \quad \Leftrightarrow \quad u = y + t$$
- If $u-b = 0$, then $u = b$ under which by optimality conditions we have
$$0 \in t\cdot[-1,1] - y + b \quad \Leftrightarrow \quad y - b \in [-t,t]$$

Now consider $g : y \mapsto t\|y-b\|_2^2$. This g is continuously differentiable so we can take its gradient directly and set it to zero, obtaining:

$$\mathbf{prox}_g(y) = \frac{y+b}{2t+1}$$

Consider now $g : (y_1,y_2) \mapsto t\gamma\|(y_1,y_2)\|_{\mathrm{iso}}$ for the iso-norm defined below. To compute the proximal operator of g, we use Moreau decomposition (7.5.8), the generalized sum rule (7.5.4), and some subdifferential calculus (7.5.2). The Fenchel conjugate of g is $g^*(y_1,y_2) = t\gamma \max_{1\le k\le n}(y_{1k}^2 + y_{2k}^2)^{1/2}$ where $y_1, y_2 \in \mathbb{R}^n$. So we have

$$\mathbf{prox}_g(y_1,y_2) = (y_1,y_2) - \mathbf{prox}_{g^*}(y_1,y_2)$$

where

$$\mathbf{prox}_{g^*}(y_1,y_2) = \arg\min_{(u,v)} \left\{ t\gamma \max_{1\le k\le n}(u_k^2+v_k^2)^{1/2} + \frac{1}{2}\|(y_1,y_2)-(u,v)\|_2^2 \right\}$$

By the generalized sum rule for the first line,

$$\partial\, \mathbf{prox}_{g^*}(y_1,y_2) = \partial\left(t\gamma \max_{1\le k\le n}(u_k^2+v_k^2)^{1/2} \right) + \partial\left(\frac{1}{2}\|(y_1,y_2)-(u,v)\|_2^2 \right)$$

$$
\begin{aligned}
&= \partial\left(t\gamma \max_{1\le k\le n}(u_k^2+v_k^2)^{1/2}\right) + \nabla\left(\frac{1}{2}\|(y_1,y_2)-(u,v)\|_2^2\right)\\
&= t\gamma\cdot \operatorname{conv}\left(\nabla((u,v)\mapsto (u_k^2+v_k^2)^{1/2})\right) + \nabla\left(\frac{1}{2}\|(y_1,y_2)-(u,v)\|_2^2\right) \quad 7.5.2\\
&= \left(\frac{t\gamma\cdot u}{(y_{1k}^2+y_{2k}^2)^{1/2}} - y_1 + u,\ \frac{t\gamma\cdot v}{(y_{1k}^2+y_{2k}^2)^{1/2}} - y_2 + v\right) \quad \text{by def. of convex hull}
\end{aligned}
$$

which vanishes when

$$
u = \left(1-\frac{t\gamma}{(y_{1k}^2+y_{2k}^2)^{1/2}}\right)y_1 \qquad v = \left(1-\frac{t\gamma}{(y_{1k}^2+y_{2k}^2)^{1/2}}\right)y_2
$$

as long as $(y_{1k}^2+y_{2k}^2)^{1/2} > 0$. If $(y_{1k}^2+y_{2k}^2)^{1/2} = 0$, then $y_{1k} = y_{2k} = 0$ so $(u,v) = (0,0)$.

For the box prox, $f : x \mapsto t\delta_S(x)$ where $S := \{x : x \in [0,1]\}$, the proximal operator is the projection. That is,

$$
\begin{aligned}
\mathbf{prox}_f(x) &= \arg\min_u \{t\delta_S(u) + \frac{1}{2}\|x-u\|_2^2\}\\
&= t\cdot \arg\min_u \{\delta_S(u) + \frac{1}{2t}\|x-u\|_2^2\}\\
&= \arg\min_{u\in S} \frac{1}{2}\|x-u\|_2^2
\end{aligned}
$$

2 Algorithms

The algorithms we present are used to solve the non-blind image deblurring problem, which is a convex optimization problem of the form

$$
\begin{aligned}
&\min_{x,y} f(x) + g(y)\\
&\text{subject to } Ax = y,
\end{aligned} \tag{7}
$$

where $f, g \in \Gamma$ have inexpensive prox-operators. We are given a vectorized, blurred image b, along with the kernel K used to blur it using the model

$$
Kx + \eta = b, \tag{8}
$$

where x is the original image, and η the added noise. Our goal is to recover the original image x. We formulate the problem in two ways, namely, one which uses the L^1 norm, and the other the L^2 norm. First,

$$
\min_x \|Kx-b\|_1 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{9}
$$

and second,

$$
\min_x \|Kx-b\|_2^2 + \delta_S(x) + \gamma\|Dx\|_{\text{iso}}, \tag{10}
$$

where $S = \{x \mid 0 \le x \le 1\}$, so that δ_S aims to restrict the range of allowed pixels. The iso norm is defined as

$$
\|(x,y)\|_{\text{iso}} = \sum_{k=1}^{n} (x_k^2 + y_k^2)^{\frac{1}{2}}
$$

2.1 Primal Douglas-Rachford splitting (Spingarn's method)

This first algorithm, which we will refer to by PDRS, was invented in the 1950s as a method to solve the heat equation [1]. Per (9), let

$$
\begin{aligned}
f(x) =& \delta_S(x)\\
g(x) =& \|Kx-b\|_1 + \gamma\|Dx\|_{\text{iso}}
\end{aligned} \tag{11}
$$

and consider the minimization problem

$$
\min_{x,y} f(x) + g(y) + \delta_0\left(\begin{bmatrix} A & -I \end{bmatrix}\begin{bmatrix} x\\ y\end{bmatrix}\right) \tag{12}
$$

with the optimality condition

$$0 \in \partial f(x) + \partial g(y) + \partial \delta_0 \left(\begin{bmatrix} A & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) =: \mathcal{F}(x,y) \tag{13}$$

As stated in the literature [2], $\mathcal{F}$ is the subdifferential of closed, convex functions, hence is a maximal monotone operator. Thus, by (5), the zeros of $\mathcal{F}$ are the fixed points of the resolvent of $\mathcal{F}$. Therefore, if we can find the fixed points of $\mathcal{F}$, we'll have found the optimal points. Unfortunately, this particular problem is too computationally expensive, so the strategy is to split $\mathcal{F}$ into two maximal monotone operators $\mathcal{A}$ and $\mathcal{B}$ which have inexpensive proximal operators. Finding the fixed points of $\mathcal{A}+\mathcal{B}$ will be equivalent to finding that of $\mathcal{F}$. Define $\mathcal{A}$ and $\mathcal{B}$ as,

$$\begin{aligned} \mathcal{A}(x,y) &= \partial f(x) + \partial g(y) = \partial(\delta_S(x)) + \partial(\|Ky-b\|_1 + \gamma\|Dy\|_{\text{iso}}) \\ \mathcal{B}(x,y) &= \partial \delta_0 \left(\begin{bmatrix} \begin{bmatrix} K \\ D \end{bmatrix} & -I \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \right) \end{aligned} \tag{14}$$

We now focus on finding the resolvents of $\mathcal{A}$ and $\mathcal{B}$,

$$\begin{aligned} J_{\mathcal{A}} = (I+\lambda\mathcal{A})^{-1} &= \mathbf{prox}_{\lambda(f+g)}(x,y) \quad \text{[by (4)]} \\ &= (\mathbf{prox}_{\lambda f}(x), \mathbf{prox}_{\lambda g}(y)) \quad \text{[by Prop. 3]} \end{aligned} \tag{15}$$

where

$$\mathbf{prox}_{\lambda f}(x) = \mathbf{prox}_{\lambda\delta_S}(x) = \arg\min_u \frac{1}{2\lambda}\|x-u\|_2^2 + \delta_S(x) = \arg\min_{u\in S} \frac{1}{2\lambda}\|x-u\|_2^2 = \mathbf{P}_S(x), \tag{16}$$

which is the euclidean projection operator onto S. It can be computed element-wise by

$$(\mathbf{P}_S(x))_i = \begin{cases} 0, & \text{if } x_i < 0 \\ x_i, & \text{if } x_i \in S \\ 1, & \text{if } x_i > 1 \end{cases} = \max\{0, \min\{x_i, 1\}\} \tag{17}$$

Now, we observe that $g(y)$ described in (11) can be written in a different form by letting

$$g(y_1,y_2,y_3) = \|y_1 - b\|_1 + \gamma\|(y_2,y_3)\|_{\text{iso}} \tag{18}$$

subject to

$$y_1 = Ky \quad \text{and} \quad \begin{bmatrix} y_2 \\ y_3 \end{bmatrix} = Dy \iff \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} K \\ D \end{bmatrix} y$$

Then, by Proposition 3, notice that the proximal operator of g may be obtained by splitting its components into

$$\varphi(y_1) = \|y_1 - b\|_1 \quad \text{and} \quad \psi(y_2,y_3) = \gamma\|(y_2,y_3)\|_{\text{iso}}$$

which gives

$$\mathbf{prox}_{\lambda g}(y_1,y_2,y_3) = (\mathbf{prox}_{\lambda\varphi}(y_1), \mathbf{prox}_{\lambda\psi}(y_2,y_3)) \tag{19}$$

We know that the proximal operator of the L^1 norm is the soft-thresholding operator T_λ, so $\mathbf{prox}_{\lambda\varphi}(y_1)$ is given element-wise as follows,

$$(\mathbf{prox}_{\lambda\varphi}(y_1))_i = (\mathbf{prox}_{\lambda\|\cdot\|_1}(y_1-b))_i = T_\lambda((y_1-b)_i) := \begin{cases} (y_1-b)_i - \lambda, & \text{if } (y_1-b)_i > \lambda \\ (y_1-b)_i + \lambda, & \text{if } (y_1-b)_i < -\lambda \\ 0, & \text{otherwise.} \end{cases} \tag{20}$$

As a result of Proposition 4, $\mathbf{prox}_{\lambda\psi}(y_2,y_3)$, is given element-wise by

$$(\mathbf{prox}_{\lambda\psi}(y_2,y_3))_i = \mathbf{prox}_{\gamma\lambda\|\cdot\|_{\text{iso}}}(y_2,y_3)_i = \alpha_i((y_2)_i,(y_3)_i),$$

where,

(provide a proof of this)

$$\alpha_i = \begin{cases} 1 - \frac{\gamma\lambda}{\sqrt{(y_2)_i^2 + (y_3)_i^2}}, & \text{if } \sqrt{(y_2)_i^2 + (y_3)_i^2} > \gamma\lambda \\ 0, & \text{otherwise.} \end{cases}$$

As for the resolvent of $\mathcal{B}$, it corresponds to the projection (x,y) of a vector $(\bar{x},\bar{y})$ onto the subspace $\{(x,y) \mid [K \quad D]^T x = y\}$, expressed as

$$J_{\mathcal{B}} = (I+\lambda\mathcal{B})^{-1} = \left(x = \left(I + [K \quad D] \begin{bmatrix} K \\ D \end{bmatrix} \right)^{-1} \left(\bar{x} + \begin{bmatrix} K \\ D \end{bmatrix} \bar{y} \right), y = \begin{bmatrix} K \\ D \end{bmatrix} x \right)$$

We now present the algorithm:

Algorithm 1: Primal Douglas-Rachford Splitting

Input : Blurred image b, step size t, relaxation parameter ρ
Output : Deblurred image x
Initialize : (z_1^0, z_2^0)

for $k = 1, 2, \ldots,$ **do**
  $x^k \leftarrow \mathbf{prox}_{tf}(z_1^{k-1})$; // Performs resolvent of $\mathcal{A}$
  $y^k \leftarrow \mathbf{prox}_{tg}(z_2^{k-1})$;
  $u^k \leftarrow (I + A^T A)^{-1}\big(2x^k - z_1^{k-1} + A^T(2y^k - z_2^{k-1})\big)$; // Performs resolvent of $\mathcal{B}$
  $v^k \leftarrow A(u^k)$;
  $z_1^k \leftarrow z_1^{k-1} + \rho(u^k - x^k)$;
  $z_2^k \leftarrow z_2^{k-1} + \rho(v^k - y^k)$;
end
return $x\mathrm{Sol} = \mathbf{prox}_{tf}(z^k)$;

2.2 Primal-Dual Douglas Rachford Splitting

In this algorithm, we consider the dual problem,

$$\text{maximize } -f^*(-A^T z) - g^*(z) \tag{21}$$

with the optimality conditions for both the primal and the dual problem

$$0 \in \begin{bmatrix} 0 & 0 & A^T & I \\ 0 & 0 & -I & 0 \\ -A & I & 0 & 0 \\ -I & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g(y) \\ 0 \\ \partial f^*(w) \end{bmatrix}$$

In the above, w is the dual variable and z is the dual multiplier for the constraint $y = Ax$, thus the dual problem is subject to

$$A^T z + w = 0$$

We reduce this to a smaller system by using the relationship $\partial g^* = (\partial g)^{-1}$, which follows from the fact that g is closed and convex, and by the properties of the subgradient and conjugate. We can eliminate y by using the above equality that $\partial g^* = (\partial g)^{-1}$, which would leave us with

$$0 \in \begin{bmatrix} 0 & A^T & I \\ -A & 0 & 0 \\ -I & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ z \\ w \end{bmatrix} + \begin{bmatrix} 0 \\ \partial g^*(z) \\ \partial f^*(w) \end{bmatrix}$$

We can further reduce this by eliminating w, using a similar equality that $\partial f = (\partial f^*)^{-1}$, we get

$$0 \in \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} + \begin{bmatrix} \partial f(x) \\ \partial g^*(z) \end{bmatrix} \tag{22}$$

We can now split the monotone operator we have above to a sum of two monotone operators given as

$$\mathcal{A}(x,z) = \begin{bmatrix} \partial f(x) \\ \partial g^*(x) \end{bmatrix} \quad \mathcal{B}(x,z) = \begin{bmatrix} 0 & A^T \\ -A & 0 \end{bmatrix} \begin{bmatrix} x \\ z \end{bmatrix} \tag{23}$$

Similar to the previous algorithm, we now find the resolvents of $\mathcal{A}$ and $\mathcal{B}$.

As seen before with the resolvent of $\mathcal{A}$, the value $(x,y) = (I + At)^{-1}(\hat{x}, \hat{z})$ is given by

$$x = \mathbf{prox}_{tf}(\hat{x}) \quad z = \mathbf{prox}_{tg^*}(\hat{z})$$

For z, we can apply moreau decomposition to get an alternate expression dealing with the conjugate g^*, thus we have,

$$z = \hat{z} - t\,\mathbf{prox}_{t^{-1}g}(\hat{z}/t)$$

Simplifying the above, we are left with,

$$z = \hat{z} - \mathbf{prox}_{gt}(\hat{z})$$

This will be a similar computation to the prox operator in algorithm 1, but we have right scalar multiplication in this case.

The resolvent for $\mathcal{B}$ is a linear mapping, so considering resolvents for skew-symmetric linear operators

$$\begin{aligned}
(I + t\mathcal{B})^{-1} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \\
&= \frac{1}{I - ((tA^T)(-tA))} \begin{bmatrix} I & -tA^T \\ tA & I \end{bmatrix} \\
&= \frac{1}{I + t^2 A^T A} \begin{bmatrix} I & -tA^T \\ tA & I \end{bmatrix}
\end{aligned}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \frac{1}{I+t^2A^TA} \begin{bmatrix} I \\ tA \end{bmatrix} \begin{bmatrix} I & -tA^T \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T$$

Letting (w,v) be the value of the resolvent of $\mathcal{B}$,

$$\begin{aligned}
\begin{bmatrix} w \\ v \end{bmatrix} &= \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \left(\begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T \right) \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I \\ -tA \end{bmatrix}^T \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} \begin{bmatrix} I & -tA^T \end{bmatrix} \begin{bmatrix} \hat{x} \\ \hat{z} \end{bmatrix} \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} (I+t^2A^TA)^{-1} (\hat{x} - tA^T\hat{z}) \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} I \\ tA \end{bmatrix} \frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)} \\
&= \begin{bmatrix} 0 \\ \hat{z} \end{bmatrix} + \begin{bmatrix} \frac{\hat{x}-tA^T\hat{z}}{(I+t^2A^TA)} \\ tA\left(\frac{\hat{x}-tA^T\hat{z}}{(I+t^2A^TA)}\right) \end{bmatrix} \\
&= \begin{bmatrix} \frac{\hat{x}-tA^T\hat{z}}{(I+t^2A^TA)} \\ \hat{z} + tA\left(\frac{\hat{x}-tA^T\hat{z}}{(I+t^2A^TA)}\right) \end{bmatrix}
\end{aligned}$$

Thus we have the solution for both $w,\ v$,

$$w = \frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)} = \frac{\hat{x}}{(I+t^2A^TA)} - \frac{tA^T\hat{z}}{(I+t^2A^TA)} \tag{24}$$

$$v = \hat{z} + tA\left(\frac{\hat{x} - tA^T\hat{z}}{(I+t^2A^TA)}\right) = \hat{z} + \frac{t\hat{A}x}{(I+t^2A^TA)} - \frac{t^2AA^T\hat{z}}{(I+t^2A^TA)} \tag{25}$$

Notice in the algorithm, we will let $\hat{x} = 2x^k - p^{k-1}$ and $\hat{z} = 2z^k - p^{k-1}$.

Here is the Primal-Dual Douglas Rachford splitting algorithm,

2.3 ADMM (Dual Douglas-Rachford)

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It can be shown that the alternating direction method of multipliers (ADMM) is exactly the Douglas-Rachford algorithm applied to the dual problem (**??**), which we rewrite as

$$\text{minimize}_z \quad f^*(-A^Tz) + g^*(z) := h(z) + k(z). \tag{26}$$

The optimality conditions for (26) are precisely $0 \in \partial h(z) + \partial k(z)$. This is a sum of two maximal monotone operators $\mathcal{A}(z) = \partial h(z)$ and $\mathcal{B}(z) = \partial k(z)$, with resolvents $\mathbf{prox}_{th}(z)$ and $\mathbf{prox}_{tk}(z)$, respectively. An equivalent formulation of the Douglas-Rachford iteration for this problem, obtained by simply interchanging the order of the updates and performing a change of variables, is thus:

$$v^k = \mathbf{prox}_{th}(z^{k-1} - w^{k-1}) \tag{27}$$

Algorithm 2: Primal-Dual Douglas-Rachford Splitting

Input : Blurred image b, step size t, relaxation parameter ρ
Output : Deblurred image x
Initialize : p^0, q^0

for $k = 1, 2, \ldots,$ **do**
  $x^k \leftarrow \mathbf{prox}_{tf}(p_1^{k-1})$; `// Performs resolvent of` $\mathcal{A}$
  $z^k \leftarrow \mathbf{prox}_{tg^*}(q^{k-1})$;
  $\begin{bmatrix} w^k \\ v^k \end{bmatrix} = \begin{bmatrix} I & tA^T \\ -tA & I \end{bmatrix}^{-1} \begin{bmatrix} 2x^k - p^{k-1} \\ 2z^k - p^{k-1} \end{bmatrix}$; `// Performs resolvent of` $\mathcal{B}$
  $p^k \leftarrow p^{k-1} + \rho(w^k - x^k)$;
  $q^k \leftarrow q^{k-1} + \rho(v^k - z^k)$;
end
return $x\mathrm{Sol} = \mathbf{prox}_{tf}(p^k)$;

$$z^k = \mathbf{prox}_{tk}(v^k + w^{k-1}) \tag{28}$$
$$w^k = w^{k-1} + \rho v^k + (1-\rho)z^{k-1} - z^k. \tag{29}$$

Proposition 5. If $h(z) = f^*(-A^T z)$, then $\mathbf{prox}_{th}(z)$ can be computed by the following:

$$\hat{x} = \arg\min_x \left(f(x) + z^T Ax + \frac{t}{2}\|Ax\|^2 \right)$$
$$\mathbf{prox}_{th}(z) = z + tA\hat{x}.$$

Proof. Notice that $\hat{x}$ as given above is optimal for

$$\begin{array}{ll} \text{minimize} & f(x) + \frac{t}{2}\|w\|_2^2 \\ \text{subject to} & w = Ax + z/t, \end{array}$$

where z is fixed and x, w are the variables. The optimality conditions for the above are $A\hat{x} + z/t = \hat{w}$, $-A^T\hat{u} \in \partial f(\hat{x})$, and $t\hat{w} = \hat{u}$, where u is the dual multiplier. Eliminating x and w and using the relationship $(\partial f)^{-1} = \partial f^*$, this is equivalent to the condition that $0 \in -A\partial f^*(-A^T\hat{u}) + \frac{1}{t}(\hat{u} - z)$, which is precisely the optimality condition that guarantees $\hat{u} = \mathbf{prox}_{th}(z) = \arg\min_u f^*(-A^T u) + \frac{t}{2}\|u - z\|_2^2$. Combining these optimality conditions also leads to the relationship $\hat{u} = z + tA\hat{x}$, which proves the proposition. □

By this proposition (applied to both h and k in (26)) the Douglas-Rachford algorithm on the dual can be rewritten as

$$\begin{aligned} \hat{x}^k &= \arg\min_x \left(f(x) + (z^{k-1} - w^{k-1})^T Ax + \frac{t}{2}\|Ax\|^2 \right) \\ v^k &= z^{k-1} - w^{k-1} + tA\hat{x}^k \\ \hat{y}^k &= \arg\min_y \left(g(y) - (v^k + w^{k-1})^T y + \frac{t}{2}\|y\|^2 \right) \\ z^k &= v^k + w^{k-1} - t\hat{y}^k \\ w^k &= w^{k-1} + \rho v^k + (1-\rho)z^{k-1} - z^k. \end{aligned}$$

If $\rho = 1$, the w update simplifies to $w^k = t\hat{y}^k$ and the entire algorithm simplifies to

$$\hat{x}^k = \arg\min_x L(x, \hat{y}^{k-1}) \tag{30}$$
$$\hat{y}^k = \arg\min_y L(\hat{x}^k, y) \tag{31}$$

$$z^k = z^{k-1} + t(A\hat{x}^k - \hat{y}^k), \tag{32}$$

where $L(x,y,z) = f(x) + g(y) + z^T(Ax-y) + \frac{t}{2}\|Ax-y\|_2^2$ is the augmented Lagrangian for the primal problem. This is the formulation of ADMM which we saw in class. To obtain the overrelaxed version for $0 < \rho < 2$, we simply replace instances of $A\hat{x}^k$ in (31) and (32) with $\rho A\hat{x}^k + (1-\rho)\hat{y}^{k-1}$.

In our implementation of ADMM for the particular problem (**??**), we first introduce a new variable u and add the constraint that $x = u$. We can then rewrite the general problem as

$$\begin{array}{ll} \text{minimize} & f(u) + g(y) \\ \text{subject to} & \begin{bmatrix} I \\ A \end{bmatrix} x + \begin{bmatrix} -I & 0 \\ 0 & -I \end{bmatrix} \begin{bmatrix} u \\ y \end{bmatrix} = 0. \end{array}$$

The augmented Lagrangian for this problem is

$$L(x,u,y,w,z) = f(u) + g(y) + w^T(x-u) + z^T(Ax-y) + \frac{t}{2}\left(\|x-u\|^2 + \|Ax-y\|^2\right).$$

We now alternate between minimizing L over x and minimizing L over (u,y). Since the problem is separable in u and y, this actually amounts to three separate minimization problems. Straightforward computations then show that the (overrelaxed) algorithm is as below (Algorithm 3).

Algorithm 3: Alternating Direction Method of Multipliers (ADMM)

Input : Blurred image b, step size t, relaxation parameter ρ
Output : Deblurred image x
Initialize : x^0, u^0, y^0, w^0, z^0

for $k = 1, 2, \ldots,$ **do**
 $x^k \leftarrow (I + A^TA)^{-1}\left(u^{k-1} + A^Ty^{k-1} - \frac{1}{t}(w^{k-1} + A^Tz^{k-1})\right)$;
 $u^k \leftarrow \operatorname{prox}_{t^{-1}f}(\rho x^k + (1-\rho)u^{k-1} + w^{k-1}/t)$;
 $y^k \leftarrow \operatorname{prox}_{t^{-1}g}(\rho Ax^k + (1-\rho)y^{k-1} + z^{k-1}/t)$;
 $w^k \leftarrow w^{k-1} + t(x^k - u^k)$;
 $z^k \leftarrow z^{k-1} + t(Ax^k - y^k)$;
end
return $x\mathrm{Sol} = (I + A^TA)^{-1}\left(u^k + A^Ty^k - \frac{1}{t}(w^k + A^Tz^k)\right)$;

2.4 La pièce de résistance: Chambolle-Pock

Acknowledgments

Please include a list of each person (with student ID) in your group and write a paragraph **for each person** with exactly what they did. Note that the Acknowledgment section does not count towards the page limit.

Example Acknowledgments would look like:

Alexandre St-Aubin (Student ID: 261115785) We ball.

Jonathan Campana (Student ID: 260985613) We ball.

Elliot Forcier-Poirier (Student ID:) We ball.

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A Appendix

Optionally include extra information (complete proofs, additional experiments and plots) in the appendix. This section will often be part of the supplemental material.